Independent AI/ML advice for Intelligence Community and Department of Defense programs

Manifold LLC is a Virginia small business run by two cleared practitioners with about ten years each inside IC acquisition and research programs. We help program offices decide whether to build or buy, and how to get the result running on accredited networks.

CORE COMPETENCIES

Strategy and roadmaps

Foundation-model strategy, capability roadmaps, build-versus-buy analysis, and technology forecasts. Technical SME on competitive acquisitions and BAA panels; AI/ML training curricula for sponsor staff.

Acquisition and compute

Technical requirements, RFP and SOW language, source-selection criteria, and vendor evaluation. GPU and TPU cluster sizing and cost models across AWS GovCloud, Azure Government, and Google Cloud's assured regions, or on-premises hardware.

Program assessment

Independent technical evaluation of AI/ML programs, maturity and risk assessment, milestone and design reviews, and sponsor-facing performance metrics.

Secure generative AI

Reference architectures for CUI at Impact Level 4 and 5, retrieval-augmented generation design and delivery, model and vendor selection in government cloud regions, evaluation plans, and bias audits.

RELEVANT EXPERIENCE OF THE PRINCIPALS

NAVY UARC · GENERATIVE AI

Designed and delivered a production retrieval-augmented generation system for CUI on Amazon Bedrock in an Impact Level 4 AWS GovCloud environment, integrated with the laboratory's directory service.

GEOSPATIAL FOUNDATION MODELS

Architected the modeling and training effort for foundation models over overhead and geospatial data, working directly with vendor engineering teams (Google, NVIDIA) on data pipelines, model design, and compute planning.

IC ACQUISITION

About ten years inside IC acquisition offices and research programs, as a Navy UARC trusted agent and as a SETA lead: requirements, source-selection and BAA evaluation, AI/ML training design, senior-leadership briefings.

DIFFERENTIATORS

- Both principals hold TS/SCI with CI polygraph
- Founding principals deliver the work themselves
- Hands-on with both acquisition and model training
- Delivered production generative AI at IL4 on GovCloud
- No competing for work we helped specify; no vendor fees

COMPANY DATA

LEGAL NAME	Manifold LLC (Virginia, formed 2026)
LOCATION	Loudoun County, Virginia
SIZE	Small business
UEI	C6P2G6PQLEB8
CAGE	23X60
NAICS	541715, 541512, 541611, 541690
CLEARANCES	TS/SCI with CI polygraph, both principals
CLASSIFIED WORK	On customer premises
WORK TYPES	Direct award, subcontract, teaming
PRINCIPALS	Paul G. Singerman, Ph.D.; John "Luke" Och

POINT OF CONTACT

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